TF-DualMethyl: Multi-Resolution Time-Frequency & Dual-Path Deep Learning for Cross-Species DNA Methylation Prediction

1. Cross-Species 4mC & 6mA Benchmarks (15 Species)

41-bp Window: Local Consensus Motif + Helical Periodicity (~10.5 bp)



Position 21 Target: C21 (4mC) / A21 (6mA)

Non-Stationary DNA Signal

- Short cis-motifs (3–7 bp)
- Helical pitch (~10.5 bp)
- Background mutation noise

15 iDNA-MS Benchmarks (489,152 Sequences Across 5 Taxonomic Clades)

4mC (4 Species)	Plants (4)	Metazoans (3)	Fungi / Prot. / Bact.
Plants & Fungi 66,938 seqs	A. thaliana, F. vesca R. chinensis, C. eq.	H. sapiens, C. elegans D. melanogaster	S. cerev., Tolyp., T. therm., Xoc BLS256

2. TF-DualMethyl: Multi-Resolution Time-Frequency & Dual-Path Architecture

Input Encoding

Token Embedding

$\sqrt{d} \cdot E(S)$, d = 128

Sinusoidal PE

Phase Relative to t=21

$X_0 \in \mathbb{R}^{41 \times 128}$

Time-Frequency Front-End

Temporal-Spatial Branch

Causal Dilated TCN (d=1,2,4)

+ Multi-Scale CNN (k=3,5,7)

3-Branch Adaptive Gated Fusion

Frequency-Wavelet Branch

Learnable rFFT VMD (K=4 Modes)

+ Haar/Db2/Gauss QMF Shrinkage

Dual-Path Backbone

2-Layer BiLSTM

5'↔3' Reading-Frame Order

$O_{lstm} \in \mathbb{R}^{41 \times 256}$

8-Head MHSA Transformer

Global All-Pairs Epistasis

$O_{trans} \in \mathbb{R}^{41 \times 128}$

SE Channel Recalibration (384-d)

Target Readout

Sequence Attention

$c = \sum \alpha_t z_t$

256-d Latent Space

5-Fold Soft-Voting

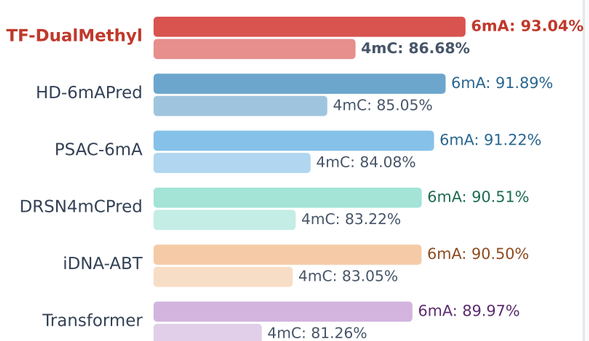
Ensemble Classifier

$P(y = 1 \mid S)$

3. State-of-the-Art Predictive Performance & Cross-Species Generalization

Macro-Average ROC-AUC (%) Across 15 Species

Outperforms 8 baselines on 15/15 datasets ($p = 3.05 \times 10^{-5}$)



Cross-Species Generalization Analysis

	A.th	F.ve	D.me	H.sa	T.th
A.th	0.96	0.95	0.90	0.91	0.56
F.ve	0.95	0.98	0.92	0.94	0.55
D.me	0.89	0.92	0.97	0.96	0.54
H.sa	0.90	0.94	0.96	0.97	0.53
T.th	0.54	0.53	0.52	0.52	0.95

Plant ↔ Metazoan

AUC: 0.933–0.975

Ciliate / Bacteria

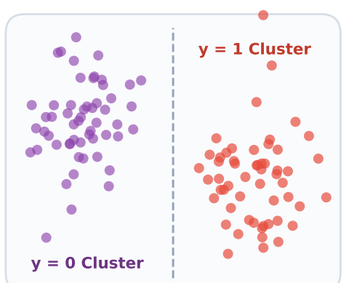
Lineage Boundary

- **Conserved eukaryotic motif grammar vs. noise**
- Evolutionary boundary in T. thermophila & Xoc
- Cold-start transfer to small-sample genomes

4. Macro-to-Micro Biological Interpretability & Conserved Motif Discovery

Feature Space (t-SNE/UMAP)

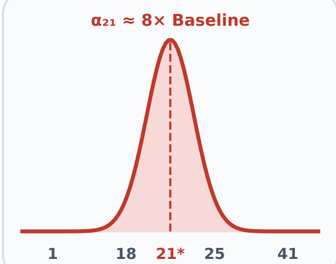
256-d Penultimate Representation



Compact Class Separation Across Diverse Genomes

41-bp Positional Attention

Autonomous Catalytic Peak at t=21



56.8%–57.7% Weight Concentrated in Central 8-bp Window (18–25)

Single-Base LIG Attribution

Conserved Catalytic Consensus Cores

A*TGCA Core

T	C	A*	T	G	C	A	C

GAGA*TTC Core

G	A	G	A*	T	T	C	A

Recovers A*T[G/T]C & [G/A]AGG Methyltransferase Target Motifs